

1.3 Hermite Interpolation

Consider the following situation: For a moving object, we know the distances traveled $d_0, d_1, \dots, d_m$, at times $t_0, t_1, \dots, t_m$, and we want a polynomial approximation of the distance function $d = d(t)$ on the entire interval containing the points $t_0, \dots, t_m$. Obviously, this is a Lagrange interpolation problem and we already know how to find the interpolation polynomial.

Now, assume that, in addition, we also know the values of the *velocities* v_i of the object at some of the times $t_i, i = \overline{0, m}$. We would expect that this additional information helps us find an *even better* approximation of the function d . However, from what we know about Lagrange interpolation, there is *no way* to include this data into our approximation. Since the velocity is the derivative with respect to time of the distance traveled, this means that we also have information about the *derivatives* of the function we want to interpolate. Maybe, at some of those times the *accelerations* are also known, in addition to the velocities. That means we have even more information, the values of the *second derivatives* of the function, at some nodes. This is a **Hermite interpolation** problem. The ideas and computational formulas are similar to the ones we used to determine the Lagrange interpolation polynomial.

1.3.1 Interpolation with double nodes

For a variety of applications, as the one described above, it is convenient to consider polynomials $P(x)$ that interpolate a function $f(x)$ and in addition have the derivative polynomial $P'(x)$ also interpolate the derivative function $f'(x)$.

Hermite interpolation problem with double nodes. Given $m + 1$ distinct nodes $x_i, i = \overline{0, m}$ and the values $f(x_i), f'(x_i)$ of an unknown function f and its derivative, find a polynomial $P(x)$ of minimum degree, satisfying the interpolation conditions

$$\begin{aligned} P(x_i) &= f(x_i), \\ P'(x_i) &= f'(x_i), \quad i = \overline{0, m}. \end{aligned} \tag{1.1}$$

Since for each node there are two values (of the function and of its derivative) given, we call them *double nodes*.

There are $2m + 2$ conditions in (1.1), so we seek a polynomial of degree (at most) $n = 2m + 1$. We determine this polynomial in a similar way to the construction of the Lagrange polynomial.

Recall, for Lagrange interpolation, the following notations:

$$\begin{aligned}\psi_m(x) &= (x - x_0) \dots (x - x_{m-1})(x - x_m), \\ l_i(x) &= \frac{(x - x_0) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_m)}{(x_i - x_0) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_m)} = \frac{\psi_m(x)}{(x - x_i)\psi'_m(x_i)},\end{aligned}\quad (1.2)$$

for $i = 0, 1, \dots, m$. Then, we expressed the Lagrange interpolation polynomial in the form

$$L_m f(x) = \sum_{i=0}^m l_i(x) f(x_i).$$

Theorem 1.1. *There is a unique polynomial $H_n f$ of degree at most n , satisfying the interpolation conditions (1.1). This polynomial can be written as*

$$H_n f(x) = \sum_{i=0}^m \left[h_{i0}(x) f(x_i) + h_{i1}(x) f'(x_i) \right], \quad (1.3)$$

where

$$\begin{aligned}h_{i0}(x) &= [1 - 2l'_i(x_i)(x - x_i)] [l_i(x)]^2, \\ h_{i1}(x) &= (x - x_i) [l_i(x)]^2, \quad i = 0, \dots, m.\end{aligned}\quad (1.4)$$

$H_n f$ is called the **Hermite interpolation polynomial** of f at the double nodes $x_0, x_1, \dots, x_m$. The functions $h_{i0}(x), h_{i1}(x)$, $i = \overline{0, m}$ are called **Hermite fundamental (basis) polynomials** associated with these points.

Proof. First we will prove that the polynomial in (1.3) *does* satisfy all interpolation conditions (i.e., existence), and then we will show that it is *the only one* to do so (i.e., uniqueness).

The degree of polynomials l_i from (1.2) is m , so the degree of h_{i0}, h_{i1} and $H_n f$ is $2m + 1 = n$.

The derivatives of the Hermite fundamental polynomials are

$$\begin{aligned}h'_{i0}(x) &= -2l'_i(x_i)(l_i(x))^2 + 2[1 - 2l'_i(x_i)(x - x_i)]l'_i(x)l_i(x), \\ h'_{i1}(x) &= (l_i(x))^2 + 2(x - x_i)l'_i(x)l_i(x).\end{aligned}$$

Notice that $l_i(x)$, $i = \overline{0, m}$ are the Lagrange fundamental polynomials, thus,

$$l_i(x_j) = \delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}.$$

Then,

$$\begin{aligned} h_{i0}(x_j) &= 0, \quad j \neq i, \\ h_{i0}(x_i) &= 1 \cdot (l_i(x_i))^2 = 1, \\ h_{i1}(x_j) &= 0, \quad j \neq i, \\ h_{i1}(x_i) &= 0. \end{aligned}$$

The values of the derivatives at the nodes are

$$\begin{aligned} h'_{i0}(x_j) &= 0, \quad j \neq i, \\ h'_{i0}(x_i) &= -2l'_i(x_i) + 2l'_i(x_i) = 0, \\ h'_{i1}(x_j) &= 0, \quad j \neq i, \\ h'_{i1}(x_i) &= 1 + 0 = 1. \end{aligned}$$

It follows that

$$\begin{aligned} (H_n f)(x_k) &= \sum_{i=0}^m \left[h_{i0}(x_k) f(x_i) + h_{i1}(x_k) f'(x_i) \right] = f(x_k), \\ (H_n f)'(x_k) &= \sum_{i=0}^m \left[h'_{i0}(x_k) f(x_i) + h'_{i1}(x_k) f'(x_i) \right] = f'(x_k), \quad k = \overline{0, m}, \end{aligned}$$

hence, the polynomial $H_n f$ given in (1.3) satisfies the interpolation conditions (1.1).

To prove uniqueness, assume there exists another polynomial G_n (of degree at most $n = 2m + 1$) satisfying relations (1.1) and consider

$$Q_n = H_n - G_n.$$

Then Q_n is also a polynomial of degree at most $n = 2m + 1$. From the interpolation conditions, it follows that

$$\begin{aligned} Q_n(x_i) &= H_n(x_i) - G_n(x_i) = f(x_i) - f(x_i) = 0, \quad i = 0, \dots, m, \\ Q'_n(x_i) &= H'_n(x_i) - G'_n(x_i) = f'(x_i) - f'(x_i) = 0, \quad i = 0, \dots, m. \end{aligned}$$

So, Q_n , a polynomial of degree at most $2m + 1$, has $m + 1$ *double* roots. By the Fundamental Theorem of Algebra, Q_n must be identically zero, thus proving the uniqueness of H_n .

□

Example 1.2. One of the most widely used form of Hermite interpolation is the **cubic** Hermite polynomial, which solves the interpolation problem with **two** double nodes $a < b$,

$$\begin{aligned} P(a) &= f(a), P(b) = f(b), \\ P'(a) &= f'(a), P'(b) = f'(b). \end{aligned} \tag{1.5}$$

Solution. First of all, let us compute the degree. The degree of the polynomial is $[2 \cdot (\text{number of nodes}) - 1]$, so, in this case,

$$n = 2 \cdot 2 - 1 = 3.$$

Letting $x_0 = a$, $x_1 = b$, with our previous notations and formulas, we have

$$\begin{aligned} \psi_1(x) &= (x - a)(x - b), \\ l_0(x) &= \frac{x - b}{a - b}, \quad l'_0(x) = \frac{1}{a - b}, \\ l_1(x) &= \frac{x - a}{b - a}, \quad l'_1(x) = \frac{1}{b - a}. \end{aligned}$$

The Hermite fundamental polynomials are given by

$$\begin{aligned} h_{00}(x) &= (1 - 2l'_0(a)(x - a))(l_0(x))^2 = \left[1 + 2\frac{x - a}{b - a}\right] \left[\frac{b - x}{b - a}\right]^2, \\ h_{10}(x) &= (1 - 2l'_1(b)(x - b))(l_1(x))^2 = \left[1 + 2\frac{b - x}{b - a}\right] \left[\frac{x - a}{b - a}\right]^2, \\ h_{01}(x) &= (x - a)(l_0(x))^2 = \frac{(x - a)(b - x)^2}{(b - a)^2}, \\ h_{11}(x) &= (x - b)(l_1(x))^2 = -\frac{(x - a)^2(b - x)}{(b - a)^2}. \end{aligned}$$

So the cubic Hermite polynomial is

$$\begin{aligned} H_3 f(x) &= \left[1 + 2\frac{x - a}{b - a}\right] \left[\frac{b - x}{b - a}\right]^2 \cdot f(a) + \left[1 + 2\frac{b - x}{b - a}\right] \left[\frac{x - a}{b - a}\right]^2 \cdot f(b) \\ &+ \frac{(x - a)(b - x)^2}{(b - a)^2} \cdot f'(a) - \frac{(x - a)^2(b - x)}{(b - a)^2} \cdot f'(b). \end{aligned}$$

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1.3.2 Newton's divided differences form

Just as in the case of Lagrange interpolation, Newton's divided differences provide a more easily computable form of the Hermite interpolation polynomial.

Consider $2m + 2$ distinct nodes $z_0, z_1, \dots, z_{2m}, z_{2m+1}$ and the Newton polynomial interpolating a function f at these nodes.

$$N_{2m+1}(x) = f(z_0) + f[z_0, z_1](x - z_0) + \dots + f[z_0, \dots, z_{2m+1}](x - z_0) \dots (x - z_{2m}),$$

with the error given by

$$R_{2m+1}(x) = f(x) - N_{2m+1}(x) = f[x, z_0, \dots, z_{2m+1}](x - z_0) \dots (x - z_{2m+1}).$$

We take the limits in the two relations above

$$z_0, z_1 \rightarrow x_0, \quad z_2, z_3 \rightarrow x_1, \quad \dots, \quad z_{2i}, z_{2i+1} \rightarrow x_i, \quad \dots \quad z_{2m}, z_{2m+1} \rightarrow x_m.$$

Denoting by $n = 2m + 1$, we get

$$\begin{aligned} N_n(x) &= f(x_0) + f[x_0, x_0](x - x_0) + f[x_0, x_0, x_1](x - x_0)^2 \\ &+ f[x_0, x_0, x_1, x_1](x - x_0)^2(x - x_1) + \dots \\ &+ f[x_0, x_0, \dots, x_m, x_m](x - x_0)^2 \dots (x - x_{m-1})^2(x - x_m) \end{aligned} \quad (1.6)$$

and for the remainder,

$$f(x) - N_n(x) = f[x, x_0, x_0, \dots, x_m, x_m](x - x_0)^2 \dots (x - x_m)^2. \quad (1.7)$$

Proposition 1.3. *Let $[a, b] \subset \mathbb{R}$ be the smallest interval containing the distinct nodes $x_0, \dots, x_m$ and $f : [a, b] \rightarrow \mathbb{R}$ be a function of class $C^{2m+2}[a, b]$. Then, for the two polynomials in (1.3) and (1.6), we have*

$$H_n f(x) = N_n(x), \forall x \in [a, b], \quad (1.8)$$

with the interpolation error

$$R_n(x) = f(x) - H_n f(x) = [\psi_m(x)]^2 \frac{f^{(n+1)}(\xi_x)}{(n+1)!}, \quad \xi_x \in (a, b). \quad (1.9)$$

Proof. By the way it was constructed (in (1.6)), obviously the polynomial N_n has degree at most n . Then, by the uniqueness of the Hermite interpolation polynomial, it suffices to show that N_n satisfies the interpolation conditions (1.1).

From (1.7), it follows that

$$f(x_i) - N_n(x_i) = 0, \quad i = 0, \dots, m.$$

Also, by the same relation, we have for the derivatives,

$$\begin{aligned} f'(x) - N'_n(x) &= (x - x_0)^2 \dots (x - x_m)^2 \frac{\partial}{\partial x} f[x, x_0, x_0, \dots, x_m, x_m] \\ &+ 2f[x, x_0, x_0, \dots, x_m, x_m] \sum_{i=0}^m \left[(x - x_i) \prod_{\substack{j=0 \\ j \neq i}}^m (x - x_j)^2 \right], \end{aligned}$$

hence,

$$f'(x_i) - N'_n(x_i) = 0, \quad i = 0, \dots, m.$$

Thus,

$$H_n f(x) = N_n(x), \quad \forall x \in [a, b]$$

and the error formula (1.9) follows directly from (1.7) and the mean-value formula for divided differences. □

Example 1.4. Let us find the polynomial and the remainder for the Hermite interpolation problem with two double nodes $a < b$, from Example 1.2.

Solution. We have

$$\begin{aligned} H_3 f(x) &= f(a) + f[a, a](x - a) + f[a, a, b](x - a)^2 \\ &+ f[a, a, b, b](x - a)^2(x - b). \end{aligned}$$

The divided differences table for two double nodes is

$z_0 = a$	$f(a)$	$\longrightarrow$	$f[a, a] = f'(a)$	$\longrightarrow$	$f[a, a, b]$	$\longrightarrow$	$f[a, a, b, b]$
		$\nearrow$		$\nearrow$		$\nearrow$	
$z_1 = a$	$f(a)$	$\longrightarrow$	$f[a, b] = \frac{f(b) - f(a)}{b - a}$	$\longrightarrow$	$f[a, b, b]$		
		$\nearrow$		$\nearrow$			
$z_2 = b$	$f(b)$	$\longrightarrow$	$f[b, b] = f'(b)$				
		$\nearrow$					
$z_3 = b$	$f(b),$						

where

$$\begin{aligned}
 f[a, a, b] &= \frac{f[a, b] - f'(a)}{b - a}, \\
 f[a, b, b] &= \frac{f'(b) - f[a, b]}{b - a}, \\
 f[a, a, b, b] &= \frac{f[a, b, b] - f[a, a, b]}{b - a} = \frac{f'(b) - 2f[a, b] + f'(a)}{(b - a)^2}.
 \end{aligned}$$

The interpolation error is given by

$$\begin{aligned}
 f(x) - H_3f(x) &= (x - a)^2(x - b)^2 f[x, a, a, b, b] \\
 &= \frac{(x - a)^2(x - b)^2}{24} f^{(4)}(\xi_x),
 \end{aligned}$$

with ξ_x belonging to the smallest interval that contains the points a, b and x .

We can find a bound for the error. Considering that on $[a, b]$, the maximum of the function $|(x - a)(x - b)|$ occurs at the midpoint of the interval, $\frac{a + b}{2}$, and that the maximum value is $\frac{(b - a)^2}{4}$, we have

$$\max_{x \in [a, b]} |f(x) - H_3f(x)| \leq \frac{(b - a)^4}{384} \max_{t \in [a, b]} |f^{(4)}(t)|.$$

■

Example 1.5 (Continuation of Example 1.1 in Lecture 4). Consider the function $f : [0.5, 5] \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}$ and the nodes $a = 1, b = 4$. Let us compare Lagrange and Hermite approximations.

Solution. For the *simple* nodes $a = 1, b = 4$, we have the interpolation conditions

$$\begin{aligned} L_1 f(1) &= f(1) = 1, \\ L_1 f(4) &= f(4) = 2, \end{aligned}$$

satisfied by the Lagrange polynomial of degree 1

$$L_1 f(x) = \frac{1}{3}x + \frac{2}{3}.$$

If the nodes are *double*, the interpolation conditions are

$$\begin{aligned} H_3 f(1) &= f(1) = 1, \\ H_3 f(4) &= f(4) = 2, \\ (H_3 f)'(1) &= f'(1) = 1/(2\sqrt{1}) = 1/2, \\ (H_3 f)'(4) &= f'(4) = 1/(2\sqrt{4}) = 1/4. \end{aligned}$$

The divided differences table is

$z_0 = 1$	$f(1) = 1$	$\longrightarrow$	$f'(1) = 1/2$	$\longrightarrow$	$f[1, 1, 4] = -1/18$	$\longrightarrow$	$f[1, 1, 4, 4] = 1/108$
		$\nearrow$		$\nearrow$		$\nearrow$	
$z_1 = 1$	$f(1) = 1$	$\longrightarrow$	$f[1, 4] = 1/3$	$\longrightarrow$	$f[1, 4, 4] = -1/36$		
		$\nearrow$		$\nearrow$			
$z_2 = 4$	$f(4) = 2$	$\longrightarrow$	$f'(4) = 1/4$				
		$\nearrow$					
$z_3 = 4$	$f(4) = 2,$						

The corresponding cubic Hermite interpolation polynomial is given by

$$H_3 f(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{18}(x-1)^2 + \frac{1}{108}(x-1)^2(x-4),$$

with derivative

$$(H_3 f)'(x) = \frac{1}{2} - \frac{1}{9}(x-1) + \frac{1}{108}(x-1)[2(x-4) + (x-1)].$$

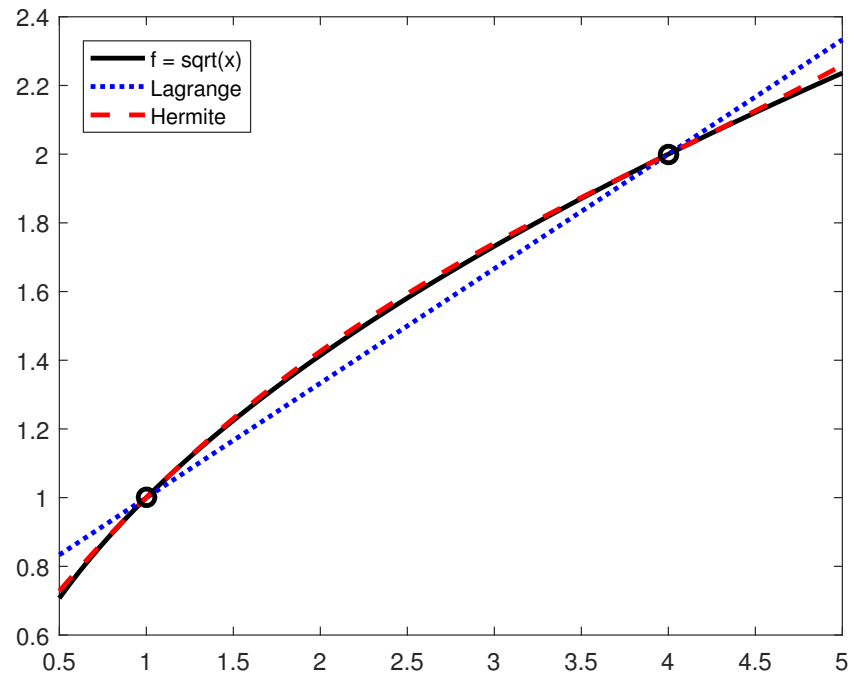


Fig. 1: Lagrange and Hermite interpolation with 2 nodes of the function $\sqrt{x}$

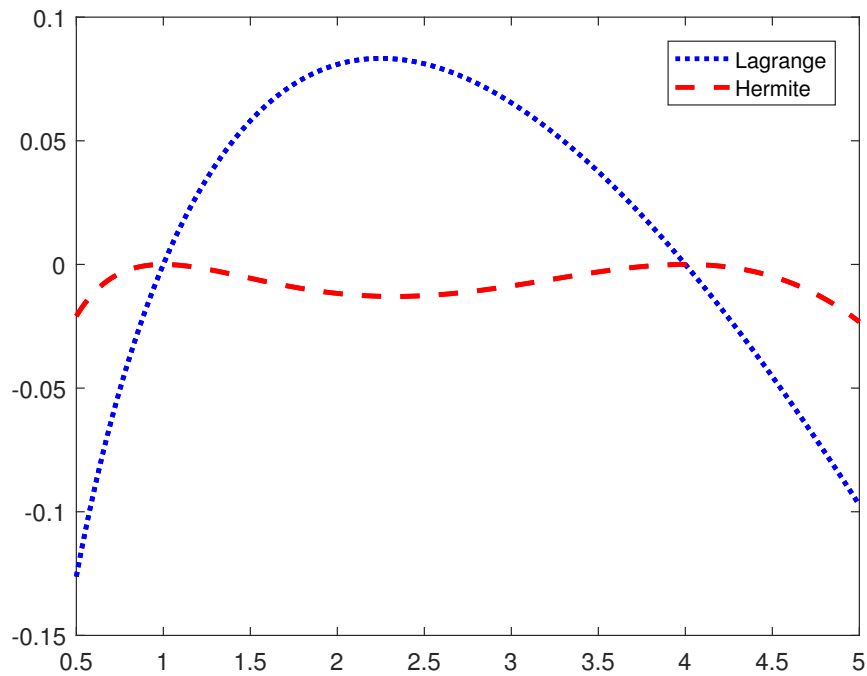


Fig. 2: Error of Lagrange and Hermite interpolation with 2 nodes of the function $\sqrt{x}$

Check that H_3f found above satisfies the interpolation conditions:

$$\begin{aligned} H_3f(1) &= 1 = f(1), \\ H_3f(4) &= 1 + \frac{3}{2} - \frac{1}{18} \cdot 9 = 2 = f(4), \\ (H_3f)'(1) &= \frac{1}{2} = f'(1), \\ (H_3f)'(4) &= \frac{1}{2} - \frac{1}{3} + \frac{1}{108} \cdot 9 = \frac{1}{4} = f'(4). \end{aligned}$$

The graphs of f and the two interpolation polynomials, L_1 , H_3 , on the interval $[0.5, 5]$, are shown in Figure 1. The interpolation errors are plotted in Figure 2.

■

1.3.3 General case

Hermite interpolation problem. Given $m + 1$ distinct nodes $x_i \in [a, b]$, $i = \overline{0, m}$,

$$\begin{aligned} x_0, & \text{ of multiplicity } r_0 + 1, \\ x_1, & \text{ of multiplicity } r_1 + 1, \\ & \dots \\ x_i, & \text{ of multiplicity } r_i + 1, \\ & \dots \\ x_m & \text{ of multiplicity } r_m + 1, \end{aligned}$$

and the values $f^{(j)}(x_i)$, $i = 0, 1, \dots, m$, $j = 0, \dots, r_i$, of an unknown function $f : [a, b] \rightarrow \mathbb{R}$ whose derivatives of order up to r_i exist at x_i , $i = \overline{0, m}$, find a polynomial $P(x)$ of minimum degree, satisfying the interpolation conditions

$$P^{(j)}(x_i) = f^{(j)}(x_i), \quad i = \overline{0, m}, \quad j = \overline{0, r_i}. \quad (1.10)$$

Above, there are

$$n + 1 \stackrel{\text{not}}{=} \sum_{i=0}^m (r_i + 1)$$

conditions, so the polynomial satisfying these relations will have degree at most n .

Theorem 1.6. *There is a unique polynomial $H_n f$ of degree at most n , satisfying the interpolation conditions (1.10). This polynomial is called the **Hermite interpolation polynomial** of the function*

f , relative to the nodes $x_0, x_1, \dots, x_m$ and the integers $r_0, r_1, \dots, r_m$, and it can be written as

$$H_n f(x) = \sum_{i=0}^m \sum_{j=0}^{r_i} h_{ij}(x) f^{(j)}(x_i). \quad (1.11)$$

Remark 1.7.

1. The functions $h_{ij}(x)$, $i = \overline{0, m}, j = \overline{0, r_i}$, are called **Hermite fundamental (basis) polynomials** and they satisfy the relations

$$\begin{aligned} h_{ij}^{(k)}(x_l) &= 0, \quad l \neq i, \quad j = \overline{0, r_i}, \quad k = \overline{0, r_l}, \\ h_{ij}^{(k)}(x_i) &= \delta_{jk}, \quad j, k = \overline{0, r_i}. \end{aligned} \quad (1.12)$$

2. If we denote by

$$\begin{aligned} u(x) &= \prod_{i=0}^m (x - x_i)^{r_i+1}, \\ u_i(x) &= \prod_{\substack{j=0 \\ j \neq i}}^m (x - x_j)^{r_j+1} = \frac{u(x)}{(x - x_i)^{r_i+1}}, \end{aligned} \quad (1.13)$$

then the fundamental polynomials $h_{ij}(x)$ in (1.11) can be written as

$$h_{ij}(x) = \frac{(x - x_i)^j}{j!} \left[\sum_{k=0}^{r_i-j} \frac{(x - x_i)^k}{k!} \left[\frac{1}{u_i(x)} \right]_{x=x_i}^{(k)} \right] u_i(x). \quad (1.14)$$

2. A more computable form can be found using Newton divided differences. Re-indexing the nodes according to their multiplicity,

$$\begin{aligned} z_0 &= x_0, \dots, z_{r_0} = x_0, \\ z_{r_0+1} &= x_1, \dots, z_{(r_0+1)+r_1} = x_1, \\ z_{(r_0+1)+(r_1+1)} &= x_2, \dots, z_{(r_0+1)+(r_1+1)+r_2} = x_2, \\ &\dots \\ z_{n-r_m} &= x_m, \dots, z_n = x_m, \end{aligned}$$

the Hermite polynomial can be written in Newton's form as

$$N_n f(x) = f(z_0) + f[z_0, z_1](x - z_0) + \dots + f[z_0, \dots, z_n](x - z_0) \dots (x - z_{n-1}), \quad (1.15)$$

with interpolation error

$$\begin{aligned} R_n(x) &= f(x) - N_n(x) = f[x, z_0, \dots, z_n](x - z_0) \dots (x - z_n) \\ &= \frac{u(x)}{(n+1)!} f^{(n+1)}(\xi_x), \quad \xi_x \in (a, b). \end{aligned} \quad (1.16)$$

Example 1.8. Consider the case of a simple node x_0 and a double node x_1 . Find the interpolant for this data and an expression for the remainder.

Solution. We have the nodes

$$\begin{aligned} x_0, \quad & \text{of multiplicity } r_0 + 1 = 1, \\ x_1, \quad & \text{of multiplicity } r_1 + 1 = 2. \end{aligned}$$

so $n + 1 = 1 + 2$ and the polynomial has degree $n = 2$.

The divided differences table:

$$\begin{array}{c|ccc} x_0 & f(x_0) & \longrightarrow & f[x_0, x_1] & \longrightarrow & \frac{f'(x_1) - f[x_0, x_1]}{x_1 - x_0} \\ & & \nearrow & & \nearrow & \\ x_1 & f(x_1) & \longrightarrow & f'(x_1) & & \\ & & \nearrow & & & \\ x_1 & f(x_1) & & & & \end{array}$$

Then,

$$\begin{aligned} H_2 f(x) &= f(x_0) + f[x_0, x_1](x - x_0) + \frac{f'(x_1) - f[x_0, x_1]}{x_1 - x_0}(x - x_0)(x - x_1) \\ &= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + \frac{f'(x_1)}{x_1 - x_0}(x - x_0)(x - x_1) \\ &\quad - \frac{f(x_1) - f(x_0)}{(x_1 - x_0)^2}(x - x_0)(x - x_1) \\ &= h_{00}f(x_0) + h_{10}f(x_1) + h_{11}f'(x_1) \end{aligned}$$

and the remainder is given by

$$R_2 f(x) = \frac{(x - x_0)(x - x_1)^2}{3!} f'''(\xi),$$

with ξ belonging to the smallest interval containing x_0 and x_1 .

Now, since H_2f has degree 2 (small), we can find it directly: we seek it of the form

$$H_2f(x) = ax^2 + bx + c$$

and determine coefficients a, b and c from the interpolation conditions:

$$\begin{cases} H_2f(x_0) = f(x_0) \\ H_2f(x_1) = f(x_1) \\ (H_2f)'(x_1) = f'(x_1) \end{cases},$$

i.e., from the linear system

$$\begin{cases} x_0^2a + x_0b + c = f(x_0) \\ x_1^2a + x_1b + c = f(x_1) \\ 2x_1a + b = f'(x_1) \end{cases}. \quad (1.17)$$

The matrix of this system,

$$V = \begin{bmatrix} x_0^2 & x_0 & 1 \\ x_1^2 & x_1 & 1 \\ 2x_1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & x_0 & x_0^2 \\ 1 & x_1 & x_1^2 \\ 0 & 1 & 2x_1 \end{bmatrix},$$

is called a *generalized Vandermonde matrix*. It is invertible and the elements of its inverse are precisely the coefficients of the fundamental polynomials h_{00}, h_{10} and h_{11} .

If the node x_0 is double and x_1 is simple, the corresponding Hermite polynomial and its error are given by

$$\begin{aligned} H_2f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f[x_0, x_1] - f'(x_0)}{x_1 - x_0}(x - x_0)^2, \\ R_2f(x) &= \frac{(x - x_0)^2(x - x_1)}{3!}f'''(\xi). \end{aligned}$$

■

Example 1.9. Find a polynomial of minimum degree that interpolates the data $f(0), f(1), f'(1)$ and $f''(1)$ (so, a *simple* node and a *triple* one). Evaluate the error.

Solution. We have the nodes

$$x_0 = 0, \text{ of multiplicity } r_0 + 1 = 1,$$

$$x_1 = 1, \text{ of multiplicity } r_1 + 1 = 3.$$

Hence, we seek the Hermite polynomial of degree at most

$$n = 1 + 3 - 1 = 3.$$

This will be of the form

$$H_3 f(x) = h_{00}(x)f(0) + h_{10}(x)f(1) + h_{11}(x)f'(1) + h_{12}(x)f''(1).$$

We compute the divided differences

$$\begin{array}{l|l}
 0 & f(0) \longrightarrow f(1) - f(0) \longrightarrow f'(1) - f(1) + f(0) \longrightarrow \frac{f''(1)}{2} - f'(1) + f(1) - f(0) \\
 & \nearrow \qquad \qquad \qquad \nearrow \qquad \qquad \qquad \nearrow \\
 1 & f(1) \longrightarrow f'(1) \longrightarrow \frac{f''(1)}{2} \\
 & \nearrow \qquad \qquad \qquad \nearrow \\
 1 & f(1) \longrightarrow f'(1) \\
 & \nearrow \\
 1 & f(1)
 \end{array}$$

Then the interpolant is

$$\begin{aligned}
 H_3 f(x) &= f(0) + (f(1) - f(0))x + (f'(1) - f(1) + f(0))x(x-1) \\
 &\quad + \left(\frac{f''(1)}{2} - f'(1) + f(1) - f(0) \right) x(x-1)^2 \\
 &= -(x-1)^3 f(0) + x(x^2 - 3x + 3)f(1) - x(x-1)(x-2)f'(1) + \frac{1}{2}x(x-1)^2 f''(1).
 \end{aligned}$$

So the fundamental polynomials are

$$\begin{aligned} h_{00}(x) &= -(x-1)^3, \\ h_{10}(x) &= x(x^2-3x+3), \\ h_{11}(x) &= -x(x-1)(x-2), \\ h_{12}(x) &= \frac{1}{2}x(x-1)^2, \end{aligned}$$

with derivatives

$$\begin{aligned} h'_{00}(x) &= -3(x-1)^2, & h''_{00}(x) &= -6(x-1), \\ h'_{10}(x) &= 3(x-1)^2, & h''_{10}(x) &= 6(x-1), \\ h'_{11}(x) &= -(3x^2-6x+2), & h''_{11}(x) &= -6(x-1), \\ h'_{12}(x) &= \frac{1}{2}(x-1)(3x-2), & h''_{12}(x) &= 3x-2. \end{aligned}$$

Now, we can better understand relations (1.12), as we can easily see that

$$\begin{aligned} \begin{cases} h_{00}(0) = 1 \\ h_{00}(1) = 0 \\ h'_{00}(1) = 0 \\ h''_{00}(1) = 0 \end{cases}, & \begin{cases} h_{10}(0) = 0 \\ h_{10}(1) = 1 \\ h'_{10}(1) = 0 \\ h''_{10}(1) = 0 \end{cases}, & \begin{cases} h_{11}(0) = 0 \\ h_{11}(1) = 0 \\ h'_{11}(1) = 1 \\ h''_{11}(1) = 0 \end{cases}, & \begin{cases} h_{12}(0) = 0 \\ h_{12}(1) = 0 \\ h'_{12}(1) = 0 \\ h''_{12}(1) = 1 \end{cases}. \end{aligned}$$

Also, it is now very easy to check that H_3f satisfies the interpolation conditions.

Alternatively, we can write the polynomial in the form

$$\begin{aligned} H_3f(x) &= \left(-f(0) + f(1) - f'(1) + \frac{1}{2}f''(1)\right)x^3 + \left(3f(0) - 3f(1) + 3f'(1) - f''(1)\right)x^2 \\ &+ \left(-3f(0) + 3f(1) - 2f'(1) + \frac{1}{2}f''(1)\right)x + f(0). \end{aligned}$$

For the remainder, we have

$$R_3f(x) = \frac{u(x)}{4!}f^{(iv)}(\xi) = \frac{x(x-1)^3}{4!}f^{(iv)}(\xi), \quad \xi \in (0, 1).$$

Now,

$$\begin{aligned} u(x) &= x(x-1)^3 = x^4 - 3x^3 + 3x^2 - x, \\ u'(x) &= 4x^3 - 9x^2 + 6x - 1 = (x-1)^2(4x-1), \end{aligned}$$

so $u(x) \leq 0$ on $[0, 1]$ and it has a local minimum at $x = \frac{1}{4}$. Thus,

$$|u(x)| \leq |u(1/4)| = \left| \frac{1}{4} \left(-\frac{3}{4} \right)^3 \right| = \frac{27}{256}.$$

Then, we find an error bound as

$$|R_3 f(x)| \leq \frac{27}{256 \cdot 4!} \max_{t \in [0,1]} |f^{(iv)}(t)| \approx 0.0044 \cdot \|f^{(iv)}\|.$$

■

Special cases

1. If all $r_i = 0, i = \overline{0, m}$, all the nodes are *simple* and we have the Lagrange interpolation formula.
2. If we consider one single node, x_0 , of multiplicity $n + 1$, the Hermite interpolation polynomial is reduced to *Taylor's polynomial*:

$$\begin{aligned} H_n f(x) &= T_n f(x) = f(x_0) + \frac{x - x_0}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \dots \\ &+ \frac{(x - x_0)^n}{n!} f^{(n)}(x_0), \end{aligned} \quad (1.18)$$

with remainder

$$R_n(f)(x) = \frac{(x - x_0)^{n+1}}{(n + 1)!} f^{(n+1)}(\xi_x). \quad (1.19)$$

3. Consider two nodes, $x_0 = a$, of multiplicity $m + 1$ and $x_1 = b$, of multiplicity $n + 1$.

The Hermite polynomial has degree

$$(m + 1) + (n + 1) - 1 = m + n + 1.$$

With the notations from Remark 1.7, we have

$$\begin{aligned} u(x) &= (x - a)^{m+1} (x - b)^{n+1}, \\ u_0(x) &= (x - b)^{n+1}, \\ u_1(x) &= (x - a)^{m+1}. \end{aligned}$$

The Hermite polynomial is of the form

$$H_{m+n+1}f(x) = \sum_{j=0}^m h_{0j}(x)f^{(j)}(a) + \sum_{i=0}^n h_{1i}(x)f^{(i)}(b) \quad (1.20)$$

and the fundamental polynomials are given by

$$\begin{aligned} h_{0j}(x) &= \frac{(x-a)^j}{j!} \left[\sum_{k=0}^{m-j} \frac{(x-a)^k}{k!} \left[\frac{1}{(x-b)^{n+1}} \right]_{x=a}^{(k)} \right] (x-b)^{n+1}, \\ h_{1i}(x) &= \frac{(x-b)^i}{i!} \left[\sum_{k=0}^{n-i} \frac{(x-b)^k}{k!} \left[\frac{1}{(x-a)^{m+1}} \right]_{x=b}^{(k)} \right] (x-a)^{m+1}. \end{aligned}$$

In Newton's form (1.15),

$$\begin{aligned} H_{m+n+1}f(x) &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(m)}(a)}{m!}(x-a)^m \\ &+ f[\underbrace{a, \dots, a}_{m+1}, b](x-a)^{m+1} + f[\underbrace{a, \dots, a}_{m+1}, b, b](x-a)^{m+1}(x-b) \\ &+ \cdots + f[\underbrace{a, \dots, a}_{m+1}, \underbrace{b, \dots, b}_{n+1}](x-a)^{m+1}(x-b)^n, \end{aligned}$$

with remainder

$$\begin{aligned} R_{m+n+1} &= f[x, \underbrace{a, \dots, a}_{m+1}, \underbrace{b, \dots, b}_{n+1}](x-a)^{m+1}(x-b)^{n+1} \\ &= \frac{f^{(m+n+2)}(\xi_x)}{(m+n+2)!}(x-a)^{m+1}(x-b)^{n+1}, \quad \xi_x \in (a, b). \end{aligned}$$